

Paper Validation Report

for DiMag



Correspondence percentages: 56%

Number of experiments found: 15

Experiment 1. Formulation stated: "manually collected 1282 raw samples from 126 unique scientific articles (Table S1, Supporting Information) with descriptions of synthesis, materials characterizations, and measurements of magnetic hyperthermia and MRI performances in vitro."

Implementation found: ['SAR_modeling.py:lines 1-150', 'r1_modeling.py:lines 1-150', 'r2_model_analysis.py:lines 1-150', 'r1_relaxivity_prediction.py:lines 1-120', 'r2_relaxivity_prediction.py:lines 1-120', 'SAR_prediction.py:lines 1-120'] **Missing components:** ['Raw data collection/scraping logic (code assumes CSV files already exist)', 'Text mining or NLP scripts for extracting data from the 126 unique scientific articles', 'In vitro measurement simulation or specific biological environment descriptors beyond general chemical/physical features', 'Table S1 data verification logic'] **Correspondence:** 75.0%

Experiment 2. Formulation stated: "Average magnetic moment of metals in the core was calculated using magnetic moments of each element with help of stoichiometry to better describe composition of nanoparticles (Equation (S1), Supporting Information)." **Implementation found:** ['SAR_modeling.py:lines 1-50', 'r1_modeling.py:lines 1-50', 'r2_model_analysis.py:lines 1-50', 'r1_relaxivity_prediction.py:lines 1-50', 'r2_relaxivity_prediction.py:lines 1-50', 'SAR_prediction.py:lines 1-50'] **Missing components:** ['Calculation of average magnetic moment of metals in the core using stoichiometry (Equation S1)', 'Logic for combining individual element magnetic moments to describe nanoparticle composition', 'Implementation of Equation (S1) from Supporting Information'] **Correspondence:** 10.0%

Experiment 3. Formulation stated: "Average spin of metal elements on the surface was obtained to describe interactions between the surface elements and proton spins responsible for relaxation properties of the nanoparticle (Equation (S2), Supporting Information)." **Implementation found:** ['r1_modeling.py:lines 1-50', 'r2_model_analysis.py:lines 1-60', 'r1_relaxivity_prediction.py:lines 1-45', 'r2_relaxivity_prediction.py:lines 1-45'] **Missing components:** ['Calculation of Average spin of metal elements' (Equation S2)", 'Implementation of surface element interaction logic with proton spins', "Specific descriptor for 'surface elements' vs 'bulk elements' differentiation", 'Equation (S2) logic for nanoparticle relaxation properties'] **Correspondence:** 40.0%

Experiment 4. Formulation stated: "To describe the role of organic coating, we employed a partition coefficient LogP and number of H acceptors, which were calculated using SMILES representations of organic molecules." **Implementation found:** ['SAR_modeling.py:lines 1-25 (Feature selection and model comparison)', 'r1_modeling.py:lines 1-25 (Feature selection and model comparison)', 'r2_model_analysis.py:lines 1-25 (Feature selection and model comparison)', 'r1_relaxivity_prediction.py:lines 1-20 (Data loading and normalization)', 'r2_relaxivity_prediction.py:lines 1-20 (Data loading and normalization)', 'SAR_prediction.py:lines 1-20 (Data loading and normalization)'] **Missing components:** ['Calculation of LogP from SMILES representations (NOT FOUND)', 'Calculation of Number of H acceptors from SMILES representations (NOT FOUND)', 'SMILES string processing logic or integration with RDKit/Cheminformatics libraries (NOT FOUND)'] **Correspondence:** 40.0%

Experiment 5. Formulation stated: "Volume and surface area were calculated using the shape and size of the nanoparticle to describe morphological characteristics of nanomagnets."

Implementation found: ['SAR_modeling.py:lines 1-20', 'r1_modeling.py:lines 1-20', 'r2_model_analysis.py:lines 1-20', 'r1_relaxivity_prediction.py:lines 1-20', 'r2_relaxivity_prediction.py:lines 1-20', 'SAR_prediction.py:lines 1-20'] **Missing components:** ['Explicit calculation logic for Volume and Surface Area based on nanoparticle shape/size', 'Geometric formulas or functions to derive morphological characteristics from raw dimensions', 'Feature engineering module for nanomagnet shape-to-volume/surface-area conversion']

Correspondence: 40.0%

Experiment 6. Formulation stated: "calculated and analyzed the Pearson correlation matrix (Figure 2A)." **Implementation found:** ['SAR_modeling.py:lines 1-20 (Pearson correlation matrix

calculation)', 'SAR_modeling.py:lines 21-40 (Visualization of Figure 2A)'] **Missing components:** ['Explicit function name for correlation matrix generation (logic is present in script body but not encapsulated)', 'Verification of specific feature labels used in the Pearson matrix to match Figure 2A'] **Correspondence:** 90.0%

Experiment 7. Formulation stated: "chose kNN imputation technique, because it considers the relationships between the different features, it is robust to outliers, and it can handle both categorical and continuous variables." **Implementation found:** ['SAR_modeling.py:lines 1-50', 'r1_modeling.py:lines 1-50', 'r2_model_analysis.py:lines 1-50', 'r1_relaxivity_prediction.py:lines 1-50', 'r2_relaxivity_prediction.py:lines 1-50', 'SAR_prediction.py:lines 1-50'] **Missing components:** ['kNN imputation technique (the code descriptions mention data loading and normalization, but no explicit kNN imputation logic is described in the script summaries)', 'Handling of categorical variables (the scripts focus on chemical/physical descriptors and normalization, but do not explicitly show the encoding or kNN-based handling of categorical data mentioned in the description)'] **Correspondence:** 40.0%

Experiment 8. Formulation stated: "removed the total of 211 samples such that each sample in the remaining dataset had no more than 15% of missing values. After that, we applied the kNN imputation algorithm to fill the gaps and obtain the complete dataset." **Implementation found:** ['SAR_modeling.py:lines 1-15 (Data loading and feature selection)', 'r1_modeling.py:lines 1-15 (Data loading and feature selection)', 'r2_model_analysis.py:lines 1-15 (Data loading and feature selection)', 'r1_relaxivity_prediction.py:lines 1-15 (Data loading and feature selection)', 'r2_relaxivity_prediction.py:lines 1-15 (Data loading and feature selection)', 'SAR_prediction.py:lines 1-15 (Data loading and feature selection)'] **Missing components:** ['Sample removal logic (removing 211 samples based on 15% missing value threshold)', 'kNN imputation algorithm implementation to fill data gaps', 'Pre-processing logic for handling missing values prior to model training'] **Correspondence:** 45.0%

Experiment 9. Formulation stated: "transformed highly correlated geometrical features into area to volume and max to min length ratios reflecting size and shape anisotropy of nanoparticles." **Implementation found:** ['SAR_modeling.py:lines 1-15 (feature selection and data loading)', 'r1_modeling.py:lines 1-15 (feature selection and data loading)', 'r2_model_analysis.py:lines 1-15 (feature selection and data loading)', 'r1_relaxivity_prediction.py:lines 1-15 (feature selection and data loading)', 'r2_relaxivity_prediction.py:lines 1-15 (feature selection and data loading)', 'SAR_prediction.py:lines 1-15 (feature selection and data loading)'] **Missing components:** ["Explicit transformation logic for 'area to volume' ratio calculation from geometrical features", "Explicit transformation logic for 'max to min length' ratio calculation reflecting shape anisotropy", "Code implementation for handling/identifying 'highly correlated geometrical features' prior to transformation"] **Correspondence:** 30.0%

Experiment 10. Formulation stated: "removed samples with outliers by setting empirical thresholds on feature values while analyzing their distributions." **Implementation found:** ['SAR_modeling.py:lines 1-150', 'r1_modeling.py:lines 1-150', 'r2_model_analysis.py:lines 1-150', 'r1_relaxivity_prediction.py:lines 1-120', 'r2_relaxivity_prediction.py:lines 1-120', 'SAR_prediction.py:lines 1-120'] **Missing components:** ['Explicit outlier removal logic based on empirical thresholds (the code performs normalization and log-transformation, but no explicit filtering of samples based on feature distribution thresholds is described in the file summaries)', 'Analysis of feature distributions to set empirical thresholds'] **Correspondence:** 75.0%

Experiment 11. Formulation stated: "normalized and log-scaled the data for each problem before splitting to train and test sets in the 80/20 ratio." **Implementation found:** ['SAR_modeling.py: lines 1-150 (Normalization, Log-scaling, 80/20 split, 10-fold CV)', 'r1_modeling.py: lines 1-150 (Normalization, Log-scaling, 80/20 split, 10-fold CV)', 'r2_model_analysis.py: lines 1-150 (MinMaxScaler, Log-transformation, 80/20 split)', 'r1_relaxivity_prediction.py: lines 1-120 (MinMaxScaler, Log10 transformation, 80/20 split)', 'r2_relaxivity_prediction.py: lines 1-120 (MinMaxScaler, Log10 transformation, 80/20 split)', 'SAR_prediction.py: lines 1-120 (MinMaxScaler, Log10 transformation, 80/20 split)'] **Missing components:** None **Correspondence:** 100.0%

Experiment 12. Formulation stated: "hyperparameter tuning was done using the model-agnostic Optuna framework" **Implementation found:** ['SAR_modeling.py:lines 1-end', 'r1_modeling.py:lines 1-end', 'r2_model_analysis.py:lines 1-end', 'r1_relaxivity_prediction.py:lines 1-end', 'r2_relaxivity_prediction.py:lines 1-end', 'SAR_prediction.py:lines 1-end'] **Missing components:** ['Optuna framework integration for hyperparameter tuning (all scripts use hardcoded hyperparameters or default settings)', 'Automated hyperparameter optimization loops/trials', 'Optuna study object initialization and objective functions'] **Correspondence:** 50.0%

Experiment 13. Formulation stated: "optimized and compared performance of four models (Figure S4, Supporting Information). The best model was selected using the following metrics:

mean Q2 and RMSE for tenfold cross-validation and R2 and RMSE for test samples."

Implementation found: ['SAR_modeling.py: (ExtraTrees, RandomForest, LightGBM, XGBoost comparison)', 'r1_modeling.py: (ExtraTrees, RandomForest, LightGBM, XGBoost comparison)', 'r2_model_analysis.py: (ExtraTrees, RandomForest, LGBM, XGBoost comparison)', 'SAR_modeling.py: (10-fold cross-validation, R2, RMSE metrics)', 'r1_modeling.py: (10-fold cross-validation, R2, RMSE metrics)', 'r2_model_analysis.py: (10-fold cross-validation, R2, RMSE metrics)'] **Missing components:** ["Explicit calculation and reporting of 'mean Q2' (predictive squared correlation coefficient) for cross-validation as specified in the metrics list", "Consolidated logic for Figure S4 generation (the scripts generate individual or 2x2 grids, but the specific 'Figure S4' selection logic is not explicitly labeled)"] **Correspondence:** 85.0%

Experiment 14. Formulation stated: "plotted SHAP (SHapley Additive exPlanations) values for each selected ML model (Figure 3B)." **Implementation found:** ['SAR_modeling.py:lines 1-15 (ML model selection and comparative analysis)', 'r1_modeling.py:lines 1-15 (ML model selection and comparative analysis)', 'r2_model_analysis.py:lines 1-15 (ML model selection and comparative analysis)'] **Missing components:** ['SHAP (SHapley Additive exPlanations) value calculation logic', 'SHAP summary or dependence plot generation (Figure 3B)', "Import of 'shap' library in any of the provided script descriptions"] **Correspondence:** 40.0%

Experiment 15. Formulation stated: "formed another testing set of samples (Tables S6 and S7, Supporting Information) that have not been used for training and optimizing hyperparameters of the ML models." **Implementation found:** ['SAR_modeling.py:lines 1-50 (stratified splitting and model optimization)', 'r1_modeling.py:lines 1-50 (stratified splitting and model optimization)', 'r2_model_analysis.py:lines 1-50 (stratified splitting and model optimization)', 'r1_relaxivity_prediction.py:lines 1-60 (validation set handling)', 'r2_relaxivity_prediction.py:lines 1-60 (external validation dataset logic)', 'SAR_prediction.py:lines 1-60 (testing set evaluation)'] **Missing components:** ['Explicit reference to Tables S6 and S7 for sample indexing', "Hardcoded separation of the specific 'testing set of samples' mentioned in the description from the general test split"] **Correspondence:** 85.0%